

Article

The Impact of AI Identity on University Students' Research Creativity and the Moderating Role of Ethical Dilemmas: An Ambidextrous Learning Perspective

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Abstract

Artificial intelligence has been effectively integrated into every stage of undergraduate research, significantly improving students' learning efficiency. Against this backdrop, artificial intelligence is no longer merely an optional external tool, but has become an extension of college students' personal capabilities—that is, "AI identity." The study constructs a moderated dual-mediation model from the perspective of ambidextrous learning. Moving beyond prior work on AI usage frequency or literacy, this study centers on AI identity and reveals the double-edged effect of AI identity on research creativity, with a positive indirect effect via exploratory learning and a negative indirect effect via exploitative learning, along with the asymmetric moderating role of ethical dilemmas on these two pathways. Using questionnaire surveys analyzing 451 college student responses, the results demonstrate that AI identity positively correlates with research creativity, where exploratory learning serves as a positive mediator while exploitative learning acts as a negative mediator. Ethical dilemmas moderate the relationship between AI identity and ambidextrous learning. These findings provide actionable insights for higher education institutions to foster students' exploratory AI use, mitigate overreliance, and establish ethical governance frameworks for AI-assisted research, thereby assisting universities in guiding students toward developing a healthy understanding of AI identity and refining ethical guidelines for AI applications.

Keywords: AI identity; ambidextrous learning; ethical dilemmas; research creativity

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1. Introduction

Artificial intelligence has begun to be applied in higher education and is gradually reshaping the future of education (Vieriu & Petrea, 2025). Current research indicates that AI tools have become indispensable aids in the learning process, with their applications spanning multiple stages, including literature reviews, data mining, hypothesis generation, and academic writing. These tools provide students with personalized learning guidance (Chu et al., 2022; Jiménez-Bucarey et al., 2021; Kamalov et al., 2023; Vieriu & Petrea, 2025). This not only optimizes traditional teaching models but also reshapes students' learning methods; AI has become a key factor influencing college students' research performance and the development

of their creativity (Singh et al., 2024; Vieriu & Petrea, 2025). Yet a growing body of research has begun to document the adverse effects of AI use, warning that overreliance on AI-generated content may erode students' capacity for critical analysis and independent problem-solving (Yan et al., 2025). Whether AI ultimately enhances or undermines college students' research creativity has thus become a contested question, with existing evidence pointing in notably different directions. A closer look suggests that studies reporting positive effects typically focus on efficiency-related outcomes, whereas studies reporting negative effects emphasize cognitive processes such as critical thinking and deep information processing. This divergence implies that the same AI tool may have opposite effects depending on which learning mechanism is activated—a possibility that has not been directly examined.

Currently, research on AI in higher education is growing rapidly. Existing studies primarily center on the effectiveness of AI tools, enhancing students' AI literacy, and exploring their acceptance of AI technology (Delcker et al., 2024; Ouyang et al., 2023; Y.-M. Wang et al., 2024; Zong & Yang, 2025). At the same time, scholars have conducted empirical studies on the impact of the digital environment on college students' learning motivation, academic performance, research capabilities, and competitiveness, thereby providing valuable insights into the relationship between the two (Montenegro-Rueda et al., 2023; Vieriu & Petrea, 2025; Webb et al., 2026). In an organizational setting, a large body of research indicates that artificial intelligence technology can reduce costs and increase productivity (Roy et al., 2025; Schwaeke et al., 2025; Webb et al., 2026; Q. Zhang et al., 2025), which also provides indirect theoretical and practical support for research into the value of AI applications in higher education settings. However, as AI becomes increasingly embedded in daily academic work, it is no longer merely an external tool that students either accept or reject. Instead, the ability to work effectively with AI is emerging as a core competency for future professionals and researchers. In a future where humans and AI will inevitably coexist, what matters is not simply whether students use AI, but how they internalize AI as part of their academic identity—that is, their AI identity. This internalization perspective remains largely overlooked in existing research, which tends to treat AI merely as an external instrument rather than a psychologically integrated aspect of the self.

AI identity refers to a cognitive state in which an individual internalizes knowledge, skills, and values related to artificial intelligence as core components of their self-concept and uses them to define themselves. Unlike AI acceptance, which reflects an individual's willingness to adopt AI tools (Albayati, 2024), or AI self-efficacy, which concerns perceived capability in using AI (J. Wang et al., 2022), AI identity captures a deeper psychological process in which AI becomes integrated into one's sense of self rather than remaining an external instrument (Carter et al., 2020). For college students, frequent use of AI-powered learning and research tools, active engagement in human-machine collaboration, or the proactive acquisition of AI-related knowledge can further stimulate and strengthen their sense of identification with AI, making it a key factor influencing their academic research behavior and the development of their creativity. Existing research indicates that AI identity can alleviate individuals' perceived identity threats posed by AI and enhance their willingness to use it, their sense of self-efficacy, and their performance at work or in their studies (Esmaeilzadeh, 2021; Mirbabaie et al., 2022; Xu et al., 2025). The strength of AI identity also influences students' utilization patterns of AI tools, thereby impacting research outcomes (Lv et al., 2022).

Nevertheless, the existing literature still has many shortcomings. On the one hand, most scholars focus solely on the positive effects of AI identity (Vieriu & Petrea, 2025), while overlooking its potential negative impacts or adverse consequences, including overreliance on AI-generated content and reduced critical thinking. This one-sided focus risks

creating an incomplete and overly optimistic understanding of AI's role in student creativity, potentially leading educators to neglect the downsides of deep AI integration and to miss opportunities for designing balanced interventions.

On the other hand, there is currently no comprehensive theoretical framework that can fully explain the complex relationship between AI identity and college students' research creativity. With advances in smart technology, artificial intelligence has increasingly permeated academic teaching and research activities, giving rise to new challenges such as overreliance and blurred boundaries regarding academic integrity. Without such a framework, it remains unclear under what conditions AI identity fosters or hinders research creativity, and how ethical concerns shape the underlying learning processes. To understand how the same AI identity can produce both beneficial and harmful effects on creativity, a theoretical framework that distinguishes between different learning mechanisms is needed.

Therefore, current research lacks a systematic theoretical explanation of how the identity of artificial intelligence influences students' creative thinking in research, and the present study aims to address this gap by proposing a moderated dual-mediation model that integrates ambidextrous learning pathways and ethical dilemmas as a key moderator.

To address the aforementioned shortcomings, this paper introduces ambidextrous learning theory for explanation. Originating from March (1991), ambidextrous learning theory categorizes human learning activities into two complementary yet competitive approaches: exploratory learning and exploitative learning. Exploratory learning involves pushing the boundaries of existing knowledge to seek out new insights, methods, and approaches; it is the driving force behind scientific innovation. Exploitative learning refers to the in-depth understanding, integration, and effective application of existing knowledge, technologies, and methods. Because AI identity may simultaneously foster exploration of new ideas and promote overreliance on automated knowledge application, ambidextrous learning theory provides a parsimonious framework capable of capturing both pathways in a single model. In the academic research context, a strong AI identity is expected to encourage exploratory learning by enabling students to push the boundaries of their research, while inhibiting exploitative learning by inducing cognitive offloading that reduces students' deep engagement with and refinement of existing knowledge. This dual-pathway framing forms the theoretical core of the present study.

In addition, ethical dilemmas may serve as a key prerequisite for AI identity to influence students' research creativity through ambidextrous learning. Ethical dilemmas refer to the various moral, integrity, privacy, and academic challenges encountered in the use of AI (Verma & Garg, 2023; Zohny et al., 2023). Specifically, students conducting research using artificial intelligence face numerous ethical challenges, such as whether AI-generated outputs qualify as original work, the legitimacy of the data used, and other issues related to academic integrity (Budhwar et al., 2023; Oniani et al., 2023).

To address the research gaps outlined above, the study proposes a moderated ambidextrous learning model to examine how AI identity influences scientific creativity.

2. Theoretical Framework and Hypotheses

2.1. AI Identity and Research Creativity

In university research environments, AI identity refers to students internalizing AI-related knowledge as a core component of their self-concept. AI identity comprises three core dimensions: dependence, emotional energy, and relatedness (Xu et al., 2025; Qin et al., 2025; Carter et al., 2020). Dependence refers to the extent to which students rely on AI to accomplish their academic tasks. Emotional energy refers to the positive feelings—such as excitement and confidence—that students experience when using AI. Relatedness refers to the degree to which students perceive AI as connected to their academic self. This

goes beyond simply “enjoying the use of AI” or “knowing how to use AI”; rather, it involves students feeling a psychological connection to AI, deriving emotional energy from it, and developing a sense of reliance on it. From the perspective of social identity theory, students with a high sense of AI identity have deeply internalized it, which influences their learning and research methods and, in turn, impacts their innovative capabilities (Faiella et al., 2025; Mikalef & Gupta, 2021).

The ambidextrous Learning theory provides important guidance for understanding these issues. It posits that people must engage in both exploratory learning and application-oriented learning in order to consistently produce high-quality innovations (March, 1991). Research demonstrates that people who are able to strike a good balance between the two tend to be more creative and innovative (Jansen et al., 2009). Therefore, any method that helps individuals achieve a balance between ambidextrous learning is likely to enhance their creativity (Limaj & Bernroider, 2019).

Strong AI identity helps lower the psychological barriers to using AI, while also giving them the confidence and ability to balance both exploratory and mastery-oriented learning approaches (Alahmad & Robert, 2020). AI identity fosters a sense of identity convergence, making individuals more willing to leverage AI for non-traditional research methods—such as interdisciplinary literature searches, formulating counterintuitive hypotheses, and exploring new analytical approaches—all of which align with the characteristics of exploratory learning. (Garbuio & Lin, 2021; van Dun et al., 2023). On the other hand, students with a strong sense of AI identity also actively use AI to improve their daily research efficiency—for example, by automating the management of references, debugging programs, and cleaning up experimental data. This frees up some of their energy from low-level, repetitive tasks, allowing them to be redirected toward more generative and exploratory intellectual work, such as formulating novel hypotheses and identifying unconventional research connections (Chen et al., 2019; Long & Liu, 2021). These two types of learning activities complement each other and work together to foster the development of innovative abilities (Chatterjee et al., 2023; Olan et al., 2022).

Research has shown that the higher people’s acceptance of new technologies, the more likely they are to view them as the most important factor in the innovation process. In research work, students with a stronger sense of AI identity are more willing to experiment with AI-based methods or explore the relationship between AI and other disciplines, thereby generating new research findings. Thus, we advance the following hypothesis:

H1. *AI identity has a positive effect on research creativity.*

2.2. The Mediating Role of Exploratory Learning

Exploratory learning involves investigating unknown areas, applying new methods, and establishing interdisciplinary connections (Jansen et al., 2006; March, 1991). The most notable feature of this learning approach is that it involves high risk, uncertainty, and a long-term commitment; therefore, it requires individuals to possess a high tolerance for risk and a strong desire to innovate (Duong, 2025; Enkel et al., 2017).

AI identity is a psychological state resulting from the internalization of technology (Carter et al., 2020), which fosters exploratory learning at both cognitive and behavioral levels. Cognitively, a positive perception of AI lowers the psychological barrier to exploring uncharted territory and reduces the perceived cost of trial and error (Alahmad & Robert, 2020). In other words, when people view artificial intelligence as an extension of their own research capabilities, they are less likely to feel psychologically inhibited when faced with new research directions and are more willing to accept the uncertainty inherent in the exploration process (Ahn, 2024). Recent research confirms that internalizing AI as part of one’s academic identity enhances confidence in tackling novel research problems

(Vieriu & Petrea, 2025). Behaviorally, AI tools facilitate exploratory learning. Research shows that large language models can rapidly generate a vast number of hypotheses, and machine learning methods can uncover hidden connections within large datasets, enabling researchers to conduct exploratory work more efficiently (Garbuio & Lin, 2021; van Dun et al., 2023). At the same time, Chen et al. (2019) argue that the integration of AI technology into creative thinking can significantly expand the scope of designers' thinking, a point that also applies to exploratory learning among researchers.

Based on ambidextrous learning theory, exploratory learning focuses on exploring unknown areas by broadening one's knowledge base, changing one's way of thinking, and trying new approaches; this serves as the foundation for producing original research findings (Gupta et al., 2006). Throughout the research process, exploratory learning not only yields tangible outcomes such as original papers and patents, but also enhances students' ability to identify problems and think across disciplines, thereby laying a solid foundation for developing their core competencies. (Jansen et al., 2009). Research also supports the view that individuals who engage in more exploratory learning perform better in terms of innovation (Limaj & Bernroider, 2019; Lin et al., 2025). Olan et al. (2022) note that the application of AI fosters innovation, thereby improving work performance. Chatterjee et al. (2023) argue that the impact of digital technology on organizational performance is mediated by individual innovative capacity. As researchers are constantly exposed to new ideas and experiment with different approaches, their thinking becomes increasingly diverse and open-minded, making it easier for them to generate novel and meaningful ideas (M. Zhang et al., 2025; Duong, 2025). From the perspective of creativity component theory, originality is the most critical element in scientific research. Only through exploratory learning can researchers conduct interdisciplinary research and apply new research methods, thereby breaking free from the constraints of traditional concepts and methods, generating entirely new ideas, and achieving corresponding results. Thus, we advance the following hypothesis:

H2. Exploratory learning mediates the relationship between AI identity and research creativity.

2.3. The Mediating Role of Exploitation Learning

Exploitation learning is a learning approach that leverages existing knowledge, processes, and outcomes to drive improvement and optimization. It helps deepen understanding of existing knowledge, increase work efficiency, and enhance the quality of output (Jansen et al., 2006; March, 1991). Compared to exploratory learning, exploitative learning involves lower risk and yields predictable results more quickly; it primarily involves the optimization and improvement in existing technologies or knowledge (Duong, 2025; Enkel et al., 2017).

AI identity may hinder exploitation learning. According to ambidextrous learning theory, human cognitive resources and time are limited, and there is a trade-off between exploratory learning and exploitative learning (Gupta et al., 2006; Duong, 2025). Students with a strong AI identity tend to rely heavily on AI-generated solutions—such as automated abstracts, data analysis frameworks, or experimental designs—which reduces their motivation to deeply process and refine existing knowledge (Lin et al., 2025; Alahmad & Robert, 2020). This cognitive offloading effect leads to less engagement with traditional research methods and a decreased likelihood of engaging in exploitative learning (S. Zhang et al., 2024). Jansen et al. (2009) also found that a high level of technological trust can lead to a path-dependence effect, whereby users place excessive faith in AI outcomes and consequently become less proactive in updating their existing knowledge. Therefore, AI identity is negatively associated with exploitative learning.

Exploitation learning has a positive impact on scientific creativity. Although exploitative learning does not lead to revolutionary innovations, it supports the development of creativity in various ways (Ahn, 2024). By improving research methodologies, such as experimental design, data analysis, and literature reviews, we can enhance efficiency and credibility, allowing researchers to focus more of their attention on creative thinking (Jansen et al., 2006; Q. Zhang et al., 2025; Duong, 2025). In addition, exploitative learning emphasizes existing knowledge and its reuse; through repeated application, it helps identify gaps or issues, thereby fostering incremental innovation (Enkel et al., 2017). High-quality exploitative learning can transform ideas that emerge during the exploratory learning process into viable and testable research findings, thereby enhancing their reliability and academic value (Limaj & Bernroider, 2019). Research has shown that exploitative learning has significant positive effects at both the individual and organizational levels (Gupta et al., 2006; Jansen et al., 2009). Thus, we advance the following hypothesis:

H3. *Exploitation learning mediates the relationship between AI identity and research creativity.*

2.4. The Moderating Role of Ethical Dilemmas

In this study, ethical dilemmas refer to students' tendency to engage in inappropriate or unethical practices when using AI technology (Verma & Garg, 2023; Singh et al., 2024). Building on the core concerns identified in prior research—such as risks of data privacy breaches, potential algorithmic biases, disputes over academic integrity, and ambiguities regarding accountability (Budhwar et al., 2023; Oniani et al., 2023; Zohny et al., 2023)—this study conceptualizes ethical dilemmas as a behavioral tendency toward technology use that conflicts with academic norms. Students with such a tendency habitually rely on shortcuts and ready-made solutions to achieve goals rather than sustained cognitive investment (Bittle & El-Gayar, 2025). Hence, ethical dilemmas can systematically alter individuals' learning strategy choices and resource allocation, thereby serving as a key moderating variable that shapes the strength and direction of the translation from AI identity into different learning behaviors.

Exploratory learning is characterized by its focus on pushing the boundaries of existing knowledge and pursuing unconventional research paths; the process is accompanied by a high probability of failure and uncertainty regarding the outcomes (Jansen et al., 2006; March, 1991). Students with higher levels of ethical dilemmas are accustomed to relying on shortcuts and ready-made AI answers, seeking quick outputs rather than genuine intellectual exploration. Their behavior is oriented toward effort avoidance and rule circumvention, rather than embracing the uncertainty of deep discovery. When such students develop a strong AI identity, they primarily use AI to obtain immediate results, which paradoxically reduces the exploratory motivation that AI identity should otherwise inspire. Consequently, the higher the level of ethical dilemmas, the weaker the positive effect of AI identity on exploratory learning. Thus, we advance the following hypothesis:

H4. *Ethical dilemmas negatively moderate the positive relationship between AI identity and exploratory learning. Specifically, the higher the level of ethical dilemmas, the weaker the facilitating effect of AI identity on exploratory learning.*

Exploitative learning focuses on the integration and optimization of existing knowledge systems, operational processes, and research findings. Its behavioral characteristics include a low-risk preference, short-term predictable returns, and a high degree of process control (Enkel et al., 2017). Exploitative learning does not require individuals to overturn existing cognitive frameworks; rather, it emphasizes achieving greater efficiency and quality improvements within established rules and established paradigms.

Students with higher levels of ethical dilemmas prefer ready-made solutions and are unwilling to engage in the elaboration and deep refinement of existing knowledge. When they form a strong AI identity, they increasingly rely on AI to perform routine tasks without substantive personal engagement, further compressing their already limited exploitative learning investment. Consequently, the higher the level of ethical dilemmas, the stronger the negative effect of AI identity on exploitative learning. Thus, we advance the following hypothesis:

H5. *Ethical dilemmas positively moderate the negative relationship between AI identity and exploitative learning. That is, ethical dilemmas strengthen the dampening effect of AI identity on exploitative learning.*

Furthermore, based on the arguments regarding the aforementioned mediating and moderating effects, this study proposes two integrated models:

H6. *Ethical dilemmas negatively moderate the mediating role of exploratory learning between AI identity and research creativity. The higher the level of ethical dilemmas, the weaker this mediating effect.*

H7. *Ethical dilemmas positively moderate the mediating role of exploitative learning between AI identity and research creativity. The higher the level of ethical dilemmas, the stronger this negative indirect effect.*

The overall research model for this paper is shown in Figure 1:

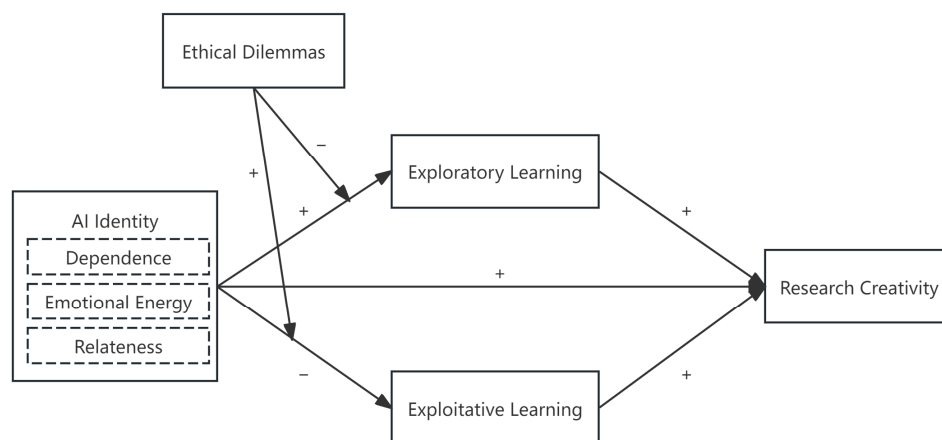


Figure 1. Model diagram.

3. Methods

3.1. Research Design and Process

Considering that university students constitute the primary user group of artificial intelligence tools and that their learning and research behaviors are significantly shaped by AI technologies (Vieriu & Petrea, 2025), the study selects currently enrolled university students as the survey population. The survey was conducted in 2026, targeting undergraduate and graduate students from universities in China. Questionnaires were distributed primarily via the WJX online survey platform and the university's online channels, including class group chats, professional exchange groups, and academic forums. In the invitation message, the purpose of the survey, its confidentiality, and the voluntary nature of participation were clearly stated. After data collection, questionnaires with incomplete responses, insufficient completion time, or uniform answer patterns were removed, yielding a final sample of 451 valid responses.

3.2. Samples and Sampling

This study used SPSS 26.0 to analyze the 451 valid questionnaires collected. In terms of gender, 251 respondents were male, accounting for 55.7% of the total, while 200 were female, representing 44.3%—a roughly balanced ratio. Regarding academic majors, the sample covered a wide range of fields. Engineering students accounted for 28.2%, followed closely by humanities and social sciences students at 27.1%. Economics and management students accounted for 23.5%, medical students for 15.3%, and students from other disciplines for 6.0%. Looking at educational level, master's students formed the largest proportion at 39.5%, followed by undergraduates at 24.6%, doctoral students at 20.2%, and a small share of students with other qualifications at 15.7%. This indicates that the study primarily focuses on master's and doctoral candidates, which aligns with the profile of students more actively engaged in research. In terms of year of study, second- and third-year students each accounted for 28.8%, fourth-year students for 27.1%, first-year students for only 10.0%, and other years for 5.8%. Thus, the study covers students from all stages, allowing insights into how research participation varies by year. The largest group—48.1%—reported using AI tools often, while 18.0% used AI tools in every learning session. Most students showed a long-term and stable habit of using AI tools, making the sample well-suited for investigating the influence of AI identity on research creativity.

3.3. Measurement

We used a well-established scale developed by previous researchers and made appropriate revisions so that it would better serve the population of university students. All items were measured on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

AI identity: This study adopted the AI identity scale developed by Carter (2013) and theoretically elaborated by Carter et al. (2015). The scale consists of 12 items covering three dimensions: relatedness, dependence, and emotional energy. An example item is, "I feel a close connection with the AI I collaborate with."

Exploratory learning: We measured exploratory learning using the 7-item scale developed by Mom et al. (2009). A typical item is, "I actively look for new entry points into my existing research direction."

Exploitative learning: This study used the 7-item scale developed by Mom et al. (2009) to measure exploitative learning. An example item is, "In my research, I mainly use methods I am very familiar with."

Research creativity: Research creativity was measured with the 6-item scale developed by Robnett et al. (2015). An example item is, "I am able to use previous ideas or findings in appropriate new ways."

Ethical dilemma: We measured ethical dilemma using the 4-item scale developed by Verma and Garg (2023). A sample item is, "To get my research work done, I use cracked versions of AI tools or software."

Control variables: Drawing on previous relevant studies and considering the characteristics of university students, we selected gender, educational level, major, year of study, and frequency of AI use as control variables to make the research findings more convincing.

4. Data Analysis and Hypothesis Testing

4.1. Confirmatory Factor Analysis and Discriminant Validity

The study employed confirmatory factor analysis. As shown in Table 1, the five-factor model adopted in the study exhibited good fit. Specifically, $\chi^2/df = 1.309$, TLI = 0.978, CFI = 0.980, GFI = 0.906, RMSEA = 0.026, and SRMR = 0.033, indicating that the model possesses good structural validity. For hypothesis testing, mediation effects were examined using Baron and Kenny's (1986) three-step hierarchical regression procedure. The

significance of indirect effects was further assessed with the bootstrapping approach recommended by Preacher and Hayes (2004), employing 5000 resamples to estimate 95% bias-corrected confidence intervals. Moderated mediation effects were tested following the frameworks of Preacher et al. (2007) and Hayes (2018).

Table 1. Confirmatory factor analysis.

Reflective Variable and Items		Mean	Cronbach's Alpha	CR	AVE	Estimate
Relatedness (RE) that AI which collaborate with me			0.856	0.856	0.598	
RE1	I am in coordination with...	4.337				0.764
RE2	I am linked with...	4.428				0.767
RE3	I am close...	4.390				0.794
RE4	I am connected...	4.421				0.768
Emotional energy (EE)			0.853	0.853	0.593	
EE1	I feel energized	4.306				0.749
EE2	I feel confident	4.450				0.755
EE3	I feel pumped up	4.330				0.797
EE4	I feel enthusiastic	4.448				0.778
Dependence (DE) that AI which collaborate with me			0.863	0.864	0.613	
DE1	I am reliant on ...	4.401				0.759
DE2	I am needing ...	4.410				0.803
DE3	I am dependent on ...	4.468				0.768
DE4	I am counting on ...	4.384				0.800
Exploratory learning (EXP)			0.911	0.912	0.596	
EL1	I actively look for new entry points for my existing research direction.	4.404				0.798
EL2	I comprehensively compare multiple research plans before making a final decision	4.424				0.768
EL3	I am committed to innovating existing research methods or content	4.353				0.764
EL4	I am willing to participate in exploratory research with uncertain outcomes	4.355				0.773
EL5	I can promptly adjust my research strategies in response to changes in my work	4.406				0.765
EL6	I take the initiative to learn entirely new skills or knowledge to support my research	4.379				0.767
EL7	I attempt to carry out research activities for which my team or university has no clear norms	4.406				0.767
Exploitation learning (EX)			0.925	0.925	0.638	
UL1	I mainly use research methods that I am very familiar with in my research	3.865				0.798
UL2	I can treat routine research tasks as daily work and complete them proficiently	3.823				0.790
UL3	I mainly use mature tools or methods to meet existing research needs	3.871				0.777
UL4	The research tasks I take on are those that I clearly know how to carry out	3.869				0.804
UL5	I prioritize the completion of short-term goals in my research	3.785				0.785
UL6	I can smoothly complete most research tasks by relying on my existing knowledge and skills	3.905				0.810
UL7	All the research activities I carry out strictly comply with the existing rules of my team and university	3.825				0.824
Research creativity (RC)			0.897	0.897	0.593	
RC1	I use previously existing ideas or work in an appropriate new way	4.346				0.743
RC2	I am very good at adapting already existing ideas or solutions	4.388				0.763
RC3	I can easily modify previously existing work processes to suit current needs	4.397				0.780
RC4	I am a good source of highly creative ideas	4.375				0.798
RC5	I demonstrate originality in teamwork	4.355				0.756
RC6	I suggest radically new ways of working.	4.461				0.779
Ethical Dilemmas (ED)			0.837	0.838	0.564	
ED1	I will use pirated AI tools or software for research and study purposes.	3.805				0.725
ED2	I will use the university's or institution's AI tools, hardware, or data resources for personal, non-research purposes	3.776				0.782
ED3	I will share the institution's AI research resources (such as data, papers, or tool access) with individuals outside the university.	3.761				0.764
ED4	I will access restricted AI research tools via the campus network or un-encrypted services	3.818				0.731

Step II: The second-order reflective construct is presented here

AI identity (AII)	0.884	0.761	0.515	
Relatedness				0.699
Emotional energy				0.751
Dependence				0.702

Source(s): Table created by authors.

4.2. Testing for Common Method Bias and Multicollinearity

Since the data for this study were collected via self-report questionnaires, common method bias (CMB) may pose a potential concern. Following the recommendations of Podsakoff et al. (2003), both procedural and statistical remedies were employed. Procedurally, the questionnaire was designed with clear instructions emphasizing confidentiality and anonymity, and the items of different constructs were separated into distinct sections to reduce consistency artifacts.

Statistically, Harman's single-factor test was conducted. All measurement items from the five constructs were entered into an exploratory factor analysis without rotation. The results showed that the first unrotated factor explained 22.182% of the total variance, which is below the 40% threshold commonly used to indicate substantive common method variance (Podsakoff et al., 2003). Therefore, common method bias does not appear to be a serious concern in this study. The corresponding results are presented in Table 2.

Table 2. Fit index table.

Component	Eigenvalue	% of Variance	Cumulative %
1	7.986	22.182	22.182
2	5.466	15.182	37.365
3	2.883	8.007	45.372
4	2.796	7.766	53.138
5	2.324	6.456	59.594
6	1.66	4.61	64.204
7	1.524	4.235	68.438

4.3. Distinctive Validity Analysis

Table 3 presents the results of the descriptive and correlational statistical analyses for each variable. The results indicate that AI identity is significantly positively correlated with exploratory learning ($r = 0.504, p < 0.01$) and significantly negatively correlated with exploitation learning ($r = -0.215, p < 0.01$), and is significantly negatively correlated with application-oriented learning ($r = -0.215, p < 0.01$). Exploratory learning is significantly positively correlated with research creativity ($r = 0.352, p < 0.01$), and exploitation learning is significantly positively correlated with research creativity ($r = 0.160, p < 0.01$). These results are consistent with the study's anticipated hypotheses and provide preliminary support for subsequent hypothesis testing.

Table 3. AVE and discriminant validity results.

Variable	RE	EE	DE	EXP	EX	RC	ED
AI identity	0.664			0.504	−0.215	0.365	−0.098
RE	0.773						
EE	0.525	0.770					
DE	0.490	0.526	0.783				
EXP	0.350	0.358	0.374	0.772			
EX	−0.193	−0.124	−0.148	−0.040	0.799		
RC	0.251	0.266	0.266	0.352	0.160	0.770	
ED	−0.058	−0.073	−0.078	−0.160	−0.207	−0.106	0.751

4.4. Hypothesis Testing

4.4.1. Main Effect Test

Following Baron and Kenny's (1986) three-step procedure, this study employed hierarchical regression to test the main effect of AI identity on research creativity, with the results shown in Table 4. AI identity has a significant positive effect on research creativity ($\beta = 0.292, p < 0.001$). Additionally, it has a significant positive effect on exploratory learning ($\beta = 0.409, p < 0.001$) and a significant negative effect on exploitation learning ($\beta = -0.172, p < 0.001$).

After including exploratory learning as a mediating variable, the positive effect of AI identity on research creativity was significant ($\beta = 0.192, p < 0.001$), and the positive effect of exploratory learning on research creativity was significant ($\beta = 0.245, p < 0.05$). Exploratory learning partially mediated the relationship between AI identity and research creativity, confirming Hypothesis 2. After including the mediating variable of exploitation learning, the positive effect of AI identity on research creativity was significant ($\beta = 0.327, p < 0.001$), and the positive effect of exploitation learning on research creativity was significant ($\beta = 0.203, p < 0.001$). Exploitation learning partially mediated the relationship between AI identity and research creativity, confirming Hypothesis 3.

Table 4. Hypothesis test results.

Variable	EXP		EX		RC		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Control variable							
Gender	0.026	0.015	0.025	0.01	0.027	0.021	0.022
Major	−0.038	−0.046	−0.003	−0.018	−0.026	−0.017	−0.026
Education Level	−0.047	−0.043	−0.003	0.004	0.053	0.065	0.054
Grade	−0.013	−0.014	0.018	0.017	0.008	0.011	0.004
Frequency of AI Use	0.022	0.026	−0.007	−0.001	0.047	0.041	0.048
independent variable							
AI identity	0.409 ***	0.407 ***	−0.172 ***	−0.178 ***	0.292 ***	0.192 ***	0.327 ***
Moderator variables							
ED		−0.102 *		−0.179 ***			
Moderating effects							
AI identity*ED		−0.095 *		−0.126 **			
Mediating variables							
EXP						0.245 ***	
EX							0.203 ***
R^2	0.169	0.191	0.031	0.086	0.090	0.14	0.116
F	15.03	13.063	2.387	5.172	7.319	10.280	9.441

Note: N = 451, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

To further validate the indirect effects, the bootstrapping method (Preacher & Hayes, 2004) was applied via IBM SPSS Statistics 27 and the PROCESS macro Version 4.2, with 5000 bootstrap resamples to generate 95% bias-corrected confidence intervals. The results are shown in Table 5. The direct effect of AI identity on research creativity was 0.282, with a Bootstrap 95% confidence interval of [0.166, 0.399], which does not include zero. The positive indirect effect of AI identity on research creativity via exploratory learning was 0.097, with a Bootstrap 95% confidence interval of [0.057, 0.139], which does not include zero; Hypothesis 2 was thus confirmed. The indirect negative effect of AI identity on research creativity via exploitation learning is −0.034, with a Bootstrap 95% confidence interval of [−0.059, −0.012], which does not include 0; Hypothesis 3 is supported.

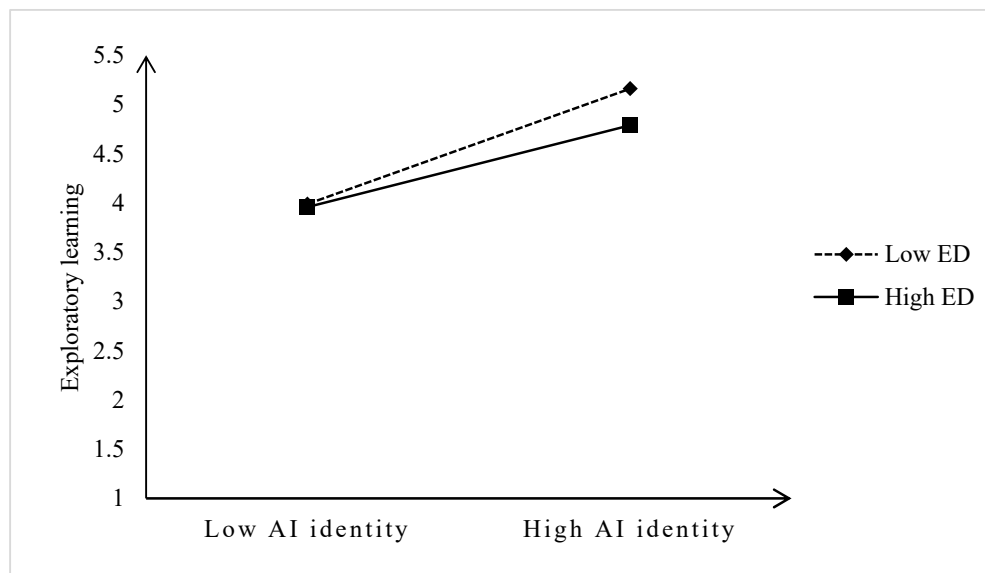
Table 5. Decomposition of the effects of AI identity on research creativity.

Path	Effect	BootSE	BootLLCI	BootULCI
Ind1: AI identity → EXP → RC	0.097	0.021	0.057	0.139
Ind2: AI identity → EX → RC	−0.034	0.012	−0.059	−0.012
Direct effect	0.282	0.059	0.166	0.399
Total effect	0.063	0.025	0.016	0.112

4.4.2. Test for Moderating Effects

The results of Model 2 in Table 4 indicate that the interaction term between AI identity and ethical dilemmas has a significant negative effect on exploratory learning ($\beta = -0.095$, $p < 0.05$), and that ethical dilemmas significantly moderate the relationship between AI identity and exploratory learning—that is, Hypothesis 4 has been confirmed. The results of Model 4 indicate that the interaction term between AI identity and ethical dilemmas has a significant moderating effect on exploitation learning ($\beta = -0.126$, $p < 0.05$). The negative coefficient signifies that ethical dilemmas strengthen the negative relationship between AI identity and exploitative learning—that is, ethical dilemmas positively moderate this negative relationship. Hypothesis 5 has been confirmed.

To illustrate the moderating effect of ethical dilemmas more intuitively, this study created simple slope plots, as shown in Figures 2 and 3. Figure 2 indicates that, compared to high levels of ethical dilemmas ($\beta = 0.071$, $p < 0.001$), the positive correlation between AI identity and exploratory learning is stronger under conditions of low ethical dilemmas ($\beta = 0.077$, $p < 0.001$), confirming Hypothesis 4. Figure 3 indicates that, compared to low levels of ethical dilemmas ($\beta = 0.087$, $p = 0.397$), the negative correlation between AI identity and exploitation learning is stronger under high levels of ethical dilemmas ($\beta = 0.081$, $p < 0.001$), confirming Hypothesis 5.

**Figure 2.** The moderating effect of ethical dilemmas on the relationship between AI identity and exploratory learning.

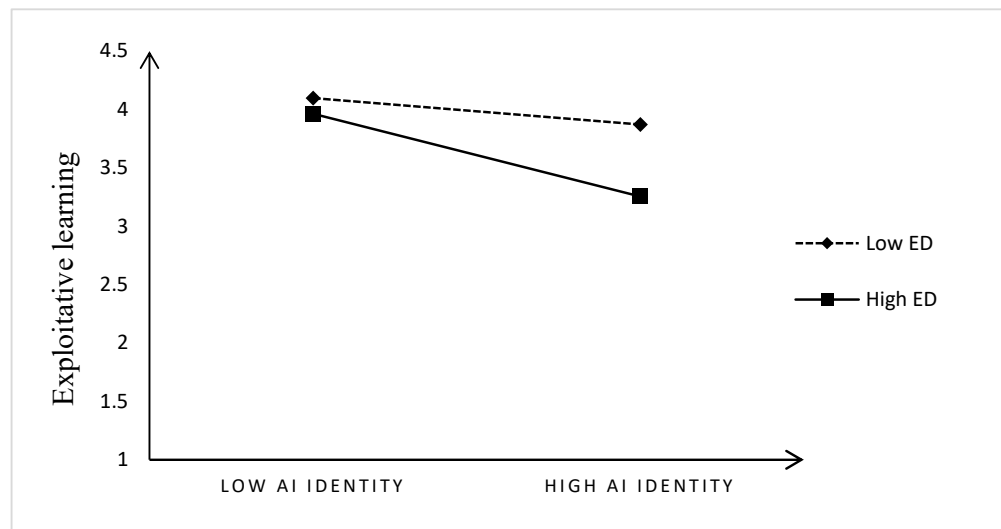


Figure 3. The moderating effect of ethical dilemmas on the relationship between AI identity and exploitative learning.

4.4.3. Testing for Moderated Mediation Effects

Following the moderated mediation framework of Preacher et al. (2007) and Hayes (2018), the study examined moderated mediation effects. As shown in Table 6, when exploratory learning served as the mediating variable, the moderated mediation effect index was -0.021 (Bootstrap 95% confidence interval: $[-0.045, -0.002]$). Specifically, at low levels of ethical dilemmas, the indirect effect of AI identity on research creativity via exploratory learning was significant (effect size = 0.150, Bootstrap 95% confidence interval $[0.090, 0.218]$, including 0). At high levels of ethical dilemmas, the indirect effect is significant (effect size = 0.095, Bootstrap 95% confidence interval $[0.045, 0.152]$), supporting Hypothesis 6. When exploitation learning serves as the mediating variable, the moderated mediation effect index is -0.023 (Bootstrap 95% confidence interval: $[-0.046, -0.005]$). At low levels of ethical dilemmas, the indirect effect of AI identity on research creativity via exploitation learning was significant (effect size = -0.014 , Bootstrap 95% confidence interval $[-0.051, -0.023]$); however, at high levels of ethical dilemmas, the indirect effect is stronger (effect size: -0.075 , Bootstrap 95% confidence interval: $[-0.125, -0.034]$), indicating that the moderated mediation holds, thereby confirming Hypothesis 7.

Table 6. Results of the mediation analysis with moderation.

Path	ED	Effect	BootSE	BootLLCI	BootULCI
AI identity → EXP → RC	Low (M-1SD)	0.150	0.032	0.090	0.218
	M	0.123	0.026	0.073	0.176
	High (M+1SD)	0.095	0.027	0.045	0.152
	AI identity*ED	-0.021	0.011	-0.045	-0.002
AI identity → EX → RC	Low (M-1SD)	-0.014	0.018	-0.051	0.023
	M	-0.044	0.015	-0.077	-0.016
	High (M+1SD)	-0.075	0.023	-0.125	-0.034
	AI identity*ED	-0.023	0.011	-0.046	-0.005

5. Discussion

Based on ambidextrous learning theory, this study developed a “double-edged sword” model of how AI identity influences scientific creativity through exploratory learning and ex-

ploitative learning, and examined the moderating role of ethical dilemmas. An empirical analysis of survey data from 451 college students revealed that AI identity has a significant positive impact on research creativity, and found that both exploratory learning and exploitative learning play a partial mediating role. On the one hand, AI identity can facilitate exploratory learning, which in turn enhances scientific creativity; on the other hand, AI identity has a negative impact on exploitative learning, yet exploitative learning can enhance scientific creativity. This constitutes a negative mediating process, and together, these two pathways form the “double-edged sword” effect of AI identity on scientific creativity. Ethical dilemmas weaken the positive relationship between AI identity and exploratory learning, yet strengthen the negative relationship between AI identity and exploitative learning; furthermore, ethical dilemmas negatively moderate the mediating effect of exploratory learning and positively moderate the mediating effect of exploitative learning. These research findings elucidate the complex relationship between AI identity and scientific creativity from the perspective of individual cognitive psychology, offering a new theoretical perspective for understanding the underlying mechanisms governing the development of scientific research capabilities among college students in the age of artificial intelligence.

5.1. Theoretical Significance

First, most existing studies treat AI technology as an objective tool, directly examining the relationship between its frequency of use or proficiency and learning outcomes (Montenegro-Rueda et al., 2023; Vieriu & Petrea, 2025), or focusing on macro-level variables such as AI literacy and technology acceptance (Bećirović et al., 2025; Shen & Cui, 2024), while relatively neglecting the critical psychological process by which individuals internalize AI as part of their self-identity (Carter et al., 2020). This paper moves beyond the perspectives of “technological objectivity” and “behavioral outcomes.” Taking AI identity as its starting point, it explores how individuals internalize artificial intelligence as part of their scientific identity, thereby giving rise to different types of exploratory and exploitative learning behaviors. This provides novel insights into psychological mechanisms underlying AI application processes, holding significant implications for AI-related research fields. By conceptualizing AI as an internalized component of academic self-concept rather than an external tool, the AI identity perspective helps reconcile seemingly contradictory findings in prior research—namely, that AI use can simultaneously yield productivity gains (Montenegro-Rueda et al., 2023; Vieriu & Petrea, 2025) and risks of deskilling and integrity erosion (Yan et al., 2025). When AI is internalized as part of one’s identity, it enables exploratory behaviors that boost creativity while also inducing cognitive offloading that undermines deep learning, thus offering a unified account of both the benefits and the costs documented in the literature.

Second, regarding the mechanism of action, this study employs the ambidextrous learning theory. Exploratory learning reflects the “positive effects” of AI identity, where a strong sense of AI identity encourages individuals to break conventions, pursue new knowledge, and generate creative outcomes; conversely, exploitative learning represents the “negative effects,” as overreliance on AI-generated standard answers weakens the ability to comprehend and reinterpret existing knowledge, thereby reducing creativity. The proposal of these “dual pathways” effectively integrates both the advantages and limitations of AI identity, facilitating a deeper understanding of the relationship between technological identity and innovative performance. Notably, the indirect effect via exploitative learning is negative in direction, forming an inconsistent mediation pathway in which the mediator suppresses rather than transmits a positive effect. This pattern reveals a hidden cost of AI identity that conventional single-pathway models would overlook.

5.2. Implications for Management

First, education administrators need to clearly recognize that a strong sense of AI identity may have significant negative effects on exploitative learning. Students who rely heavily on AI may be more inclined to rely on algorithm-generated results, while neglecting to thoroughly study, thoroughly digest, and critically reflect on existing bodies of knowledge. If left unchecked, this trend could lead to cognitive depth in the learning process being diluted by technological efficiency, thereby undermining students' solid grasp of classic theories and fundamental methodologies in the discipline. Therefore, in the design and organization of university-level courses, we must steadfastly prioritize the teaching of foundational theories and classical methods. Educators should guide students in developing a balanced perspective on AI usage—that is, while leveraging AI to enhance information processing efficiency, they should always maintain a healthy sense of respect and critical scrutiny toward traditional deep learning methods, and continue to hone their abilities in logical reasoning and knowledge internalization independent of algorithmic assistance. Only in this way can we strike a balance between technological empowerment and deepening understanding.

Second, universities urgently need to establish clear, well-defined, and enforceable policies governing AI use in academic research. The moderation analysis in this study indicates that high levels of ethical dilemmas significantly weaken the positive effect of AI identity on exploratory learning. This suggests that ethical constraints that appear overly strict or vague may, by exacerbating individuals' confusion and anxiety, become a hidden institutional cost that stifles creativity. Concretely, institutions should require students to include an AI-use disclosure statement when submitting coursework or academic papers, specifying which parts of the research process involved AI assistance and to what extent. Meanwhile, specific provisions addressing AI should be added to academic integrity codes, clearly distinguishing between legitimate AI-assisted scholarship and academic misconduct with discipline-specific examples and corresponding consequences. In terms of assessment, oral defenses and process-oriented evaluation should carry greater weight, with process materials such as research proposals, literature notes, and data analysis scripts incorporated into the grading system rather than evaluating only the final submitted text. Regarding AI literacy training, universities should integrate relevant content into existing information literacy courses, helping students learn to verify the accuracy of AI-generated information, identify potential algorithmic biases in outputs, and develop the habit of maintaining human oversight at critical stages of research. These measures are not isolated initiatives but function collectively: only when disclosure mechanisms make usage transparent, acceptable-use policies clarify behavioral boundaries, assessment design enables genuine intellectual engagement to be recognized, and literacy training equips students with critical judgment can ethical guidelines shift from an external constraint that induces anxiety to an internal scaffold that provides security, enabling students to pursue innovative research with confidence under the protection of well-defined norms.

Third, universities should make fostering a healthy AI identity among students a key educational objective, rather than limiting their focus to training in AI tool operation. The empirical analysis indicates that AI identity can significantly enhance scientific creativity through the mediating mechanism of exploratory learning. The practical implication is that instructional intervention should aim at facilitating a structural shift in students' self-concept—helping them internalize AI as part of their academic identity rather than treating it as an external tool. Universities may consider establishing a curriculum that deeply integrates AI with disciplinary innovation, or guiding students to actively participate in human-machine collaborative research projects, thereby enabling them to continuously receive positive feedback from technological collaboration throughout the process of genuine academic inquiry. When students repeatedly experience the increased efficiency and

sense of validation that comes from recognizing AI capabilities as a core component of their self-identity, their psychological barriers and defensive attitudes toward emerging technologies will gradually dissolve, enabling them to consciously extend the empowering effects of AI to the breakthrough and reconstruction of traditional research paradigms.

5.3. Limitations and Future Directions

Due to limitations in research conditions and the study design, this study still has several shortcomings that require further refinement and breakthroughs in future research. First, all of the sample data in this study were collected from a population of college students, and the core variables were measured using self-report questionnaires. Although self-report scales are widely recognized in social science research as an effective means of capturing individual psychological constructs, their reliance on a single data source makes it difficult to completely avoid the potential interference of common method bias. Future research is advised to incorporate a multi-source evaluation mechanism during the data collection phase; for example, by inviting supervisors or peers to provide external evaluations of participants' research creativity, or by scraping behavioral log data from research platforms to obtain objective behavioral indicators, separation of variable sources at the measurement level can be achieved.

Second, this study employed a cross-sectional research design, which involves collecting data on all variables at a single point in time and using this data to infer the structural relationships among the variables. Although this design can to some extent reflect the patterns of covariation among variables, it cannot provide rigorous temporal evidence to establish a causal relationship. Future research should employ longitudinal designs to examine the temporal sequence of effects among variables. In addition, the incorporation of experimental or quasi-experimental studies would also be of significant value. Researchers could design interventions to elicit AI identity, using situational experiments to induce varying degrees of state-level AI identity in participants, and then observe differences in their performance on subsequent creative tasks, thereby providing more robust empirical support for causal inferences.

Third, while this study focuses on the moderating role of ethical dilemmas as a contextual factor, the scope of AI identity's influence on scientific creativity is far too broad to be fully captured by a single variable.

In addition to individuals' perceptions of ethical risks, the degree of dynamism in the external environment and the level of transparency inherent in AI systems may both exert significant moderating effects on the aforementioned relationship. Future research could incorporate these variables into the analytical framework to construct a more comprehensive moderating model, thereby systematically revealing the boundary conditions under which AI identity functions effectively across different technological and contextual conditions.

Fourth, this study treated exploratory learning and exploitative learning as parallel mediators and did not examine their potential balance or synergistic effects on research creativity. Previous research suggests that creativity often benefits from an appropriate balance between exploration and exploitation. Future research could extend our model by incorporating a balance index (e.g., exploratory learning–exploitative learning) or an interaction term to test whether ambidextrous synergy explains additional variance in research creativity beyond the two independent pathways. Such an approach would provide a more nuanced understanding of how AI identity influences creativity through the interplay of different learning modes.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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